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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	The Powerpuff girls
Selected Domain	Domain 2
Selected SDG Track (1–17)	Decent work and economic growth
Project / Solution Name	Billguard
Name (Leader)	Aisuluu Abdullaeva
Email & Phone (Leader)	+996552577577
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian SMEs must follow the LHDN's e-invoice regulations and rules, which can cause issues when businesses buy from foreign suppliers or local suppliers that cannot issue e-invoices. The clerks retype Chinese, Malay, or handwritten bills and have to track the TIN, exchange rates, and SST for submitting the self-issued invoice into the system.

All micro and small business owners are affected by that, including their staff and bookkeepers. Affected firms rarely have an in-house tax specialist, and experienced accounting staff are increasingly expensive.

Impact is caused in the following areas:

1. Economic: errors delay records and create compliance risk and cash-flow stress.
2. Social: Language and digital-skills gaps leave the small operators in disadvantage position to comply, increasing the gap with larger firms.

BillGuard reads the bill, flags problems in plain language, requires role-based human approval, and queues submissions safely, keeping people in control and every action logged.

Summary: SDG 8.2 and SDG 8.3

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Billguard is a web application that turns a supplier bill into a checked, approved, queued self-build e-invoice

1. A staff member uploads a photo or PDF of a bill or pastes its text
2. AI scans it and validate it for missing TIN, wrong SST, total amount, foreign currency handling, duplicates
3. Based on the missing parts, it suggests solution for each detected problem

4. The system routes it to the right approver (accountant/owner) by role, and blocks people from approving their own entries.
5. Approved invoices enter a queue that throttles submissions, retries after certain period of time, and never creates duplicates.

Each step is recorded. For the demo version, the tax portal is simulated and all data is synthetic.

AI application: a vision-language model (vLLM), called through a GenAI, reads Chinese, Malay and English and handwritten bills (haven't tested handwritten and malay). We keep validation, routing, and approval rule-based so decisions are clear and reasonable. A person reviews the AI's conclusion before proceeding with approval/rejection, and system never invents new values.

Tools:

1. GenAI APIs: calling a hosted vision model through an OpenAI-compatible API keys, with retries and fallbacks.
2. SQLite for invoices and data tables (data can move to PostgreSQL)
3. A serverless GPU endpoint for the model and Replit for the web app.
4. Python with FastAPI, role-based access control.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

1. Python 3
2. FastAPI
3. Pydantic
4. HTML/JV

AI & API:

1. A vision – language model, Qwen2.5-VL (SMTH)

Database: SQLite for invoices

Hosting:

1. Replit for web app
2. Runpod for model (Qwen2.5-VL)
3. Pytest

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We began with journey of:

Supplier -> staff -> owner

We kept the logic as simple as possible and decided to use AI for improving the efficiency of the practices used in small business right now. We tried to adapt it to the different industries to make it applicable for different types of businesses. Our system aims to reduce the pressure of constant tracking, lack of skilled labor, and provide the owners with clear system for sorting the invoices. We built a working version with a single-screen interface, plain-language error messages and role-based approval.

Right now, we are strengthening the parts that we didn’t had an opportunity to improve, such as the identity provider, and a production-grade database. Further features implementations include the financials reports built from the validated invoices. We are planning to test with real SME users, measure OCR accuracy and connect to LHDN’s sandbox on place of the simulated portal.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8, Target 8.2: higher economic productivity through technological upgrading and innovation. Target 8.3: support for entrepreneurship and the growth of micro, small and medium-sized enterprises. Our product removes manual and error prone invoice checking so small businesses can spend that time on running and growing the business. Secondarily supports, SDG 9, supporting digital infrastructure for small firms.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Manual handling of one supplier invoice takes approximately 12 minutes of opening, reading, typing in, checking and filing of data, and manual data entry carries an error rate in the range of 1 to 4%. Sources: [Invoice Data Extraction, Invoice Processing Time Benchmarks]; [Lido, The True Cost of Manual Invoice Processing, March 2026]
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Processing time per invoice from upload time to approved record. % detection of deliberately planted errors flagged before approval against manual time of detection.
Phase 2 Impact Target	A 50% reduction in the processing time per invoice from approximately 12 minutes to about 6 minutes or less with

Metric Category	Your Response
Your realistic, percentage-based target by the final prototype presentation *e.g., "20% improvement in computational resource use."*	automation, and 99% of planted errors flagged before approval, measured on synthetic invoices.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☒ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☒ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Claud AI	Code	Open-source
Replit	Code	Open-source

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date] 24/10/2026